



Diesel Engine Emissions & Exhaust Gas After Treatment

PowerGen Diesel

TSLs / TRA

April 2022

This PowerPoint presentation is protected by copyright Rolls-Royce Solutions GmbH expressly reserves all rights to this presentation. Publication, duplication or disclosure to third parties – even in the form of extracts – are strictly forbidden unless expressly approved by the Management of Rolls-Royce Solutions GmbH. Rolls-Royce Solutions GmbH furthermore reserves all rights, particularly in regard of the use, processing reproduction of content related to any intellectual property claims.





Contents

- 01 Introduction / Disclaimer
- 02 Motivation for Diesel engine emission reduction
- 03 Diesel engine Emission regulation
- 04 Diesel Engine Technologies
- 05 *mtu* Engine Technologies



01

Introduction / Disclaimer



Chapter 1

Introduction / Disclaimer

Introduction	5
Disclaimer	6



Introduction

Typical diesel engine exhaust emissions are determined by diesel combustion process which transforms the fuel's chemical energy into usable mechanical energy. Besides CO₂, which theoretically speaking would be the combustion product in an ideal chemical reaction, various other combustion by-products develop in small quantities during combustion. In past decades, several of these combustion by-products have increasingly become the focus of attention due to their hazardous effect on health. As a result, they have been constantly subject to increasing regulations. This development in the form of increasingly stringent emission standards is expected to continue in future.

In its role as a leader in high-speed diesel engine technology for industrial applications, *mtu* Takes its responsibility towards people and engine environment extremely seriously. Consequently, the company has become a leading competitor with its in-house development Of high-tech solutions in the field of exhaust emissions.

More and more strict emission limits, like the American EPA Tier4 regulation require new and innovative technical solutions. In addition to precise coordination of all the relevant engine sub-assemblies such as injection system, engine management or components which control gas exchange, some means of treating the exhaust gas after it has left the engine is necessary. Although several different technologies are available, these can only produce ideal results if they are developed together with the basic engine and the appropriate exhaust aftertreatment system to create an optimum solution in all areas.



Disclaimer

The information in this document is considered as guideline and for information only. Please verify with your authorities in charge based on your specific projects.

The information is given in good faith based upon the latest information available to Rolls-Royce Solutions GmbH, no warranty or representation is given concerning such information, which must not be taken as establishing any contractual or other commitment binding upon Rolls-Royce Solutions GmbH or any of its subsidiary or associated companies.



02

Motivation for Diesel engine emission reduction



Chapter 2

Motivation for Diesel engine emission reduction

Oxides of nitrogen, hydrocarbon	9
Carbon monoxide, particulate matter	10
Formaldehyde, sulfur compounds	11

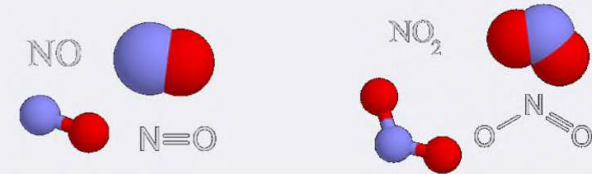


Oxides of nitrogen, hydrocarbon

Harmful diesel exhaust gas emission components and their effects

In unnaturally high concentrations the following effects may occur:

Oxides of nitrogen (NO_x)

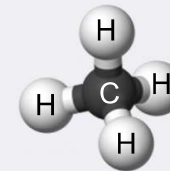


Acid former by contact with H_2O : Acid rain, irritating / toxic to mucous membrane

Greenhouse gas: Global warming

Ozone killer: Harmful to creatures

→ Hydrocarbon (HC)



Highly toxic volatile organic compound (to all terrestrial organisms): damage to the central nervous system, some cardiogenic, irritating mucous membrane, obstruction of oxygen absorption in the blood

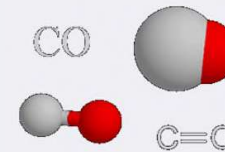


Carbon monoxide, particulate matter

Harmful diesel exhaust gas emission components and their effects

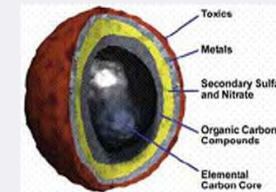
In unnaturally high concentrations the following effects may occur:

→ Carbon monoxide (CO)



Low concentration: Damage to the central nervous system, damage to cardiovascular system, High concentration: Suffocation by blocking oxygen transport in the blood

→ Particulate matter (PM)



According to World Health organization, PM is considered to be Cardiogenic